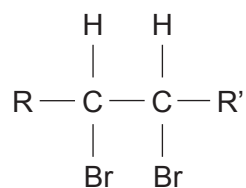


8. (a) Compound **M** is an aliphatic compound, which contains two bromine atoms.



compound **M**

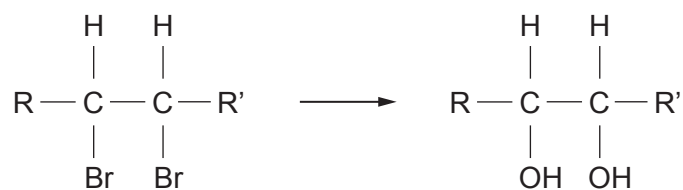
- (i) Compound **M** reacts with an excess of alcoholic potassium hydroxide solution to give compound **N**, which has the formula $\text{R}-\text{C}\equiv\text{C}-\text{R}'$.

Suggest the role of the alcoholic potassium hydroxide in this elimination reaction.

[1]

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- (ii) Compound **M** reacts with aqueous sodium hydroxide to give compound **P**.



compound **P**

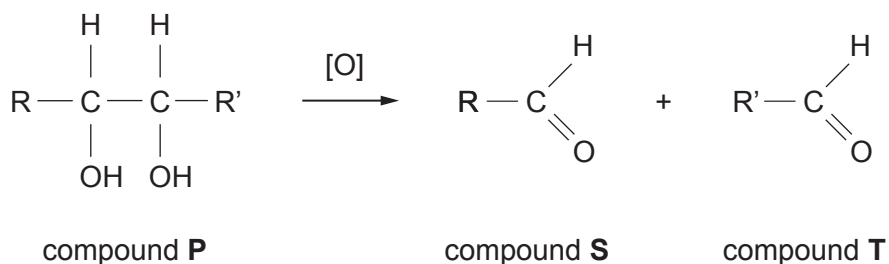
State the type of reaction mechanism occurring.

[1]

.....



- (iii) Compound **P** can undergo bond fission when it is heated with a suitable oxidising agent to form two aldehydes **S** and **T**.



The mass spectrum of compound **S** showed a molecular ion at m/z 72.

- I. The mass spectrum of compound **S** showed that the alkyl group R was branched.

Deduce the structure of compound **S**. Show your reasoning. [2]

- II. Compounds **S** and **T** both reacted with 2,4-dinitrophenylhydrazine to give solid derivatives.

State the colour of these derivatives. [1]

.....

- III. The melting temperatures of the derivatives of compounds **S** and **T** in part II above are shown in the table.

Compound	Melting temperature/ $^{\circ}\text{C}$
S	185–187
T	186–189

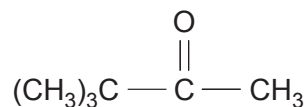
Describe how the melting temperature of a 50:50 mixture of the two derivatives differs from those shown in the table. [1]

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.....



- (b) Pinacolone is prepared from a diol containing 6 carbon atoms.



pinacolone

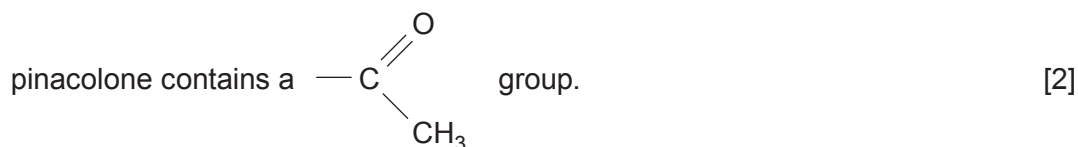
After the reaction, flammable pinacolone is distilled from the reaction mixture. The distillate consists of two layers – an aqueous layer and a less dense layer containing mainly pinacolone.

- (i) Describe how you would separate these two layers. [1]

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.....

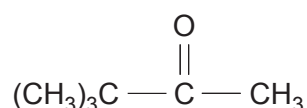
- (ii) Describe a chemical test, giving reagent(s) and observations, to show that



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- (iii) The high resolution ^1H NMR spectrum of pinacolone consists of two signals.



pinacolone

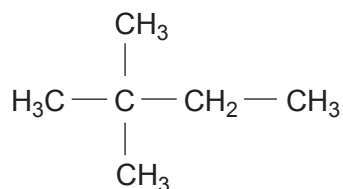
Complete the table below that describes these two signals. [2]

Protons	Splitting pattern	Relative peak area
CH_3CO		
$(\text{CH}_3)_3\text{C}$		

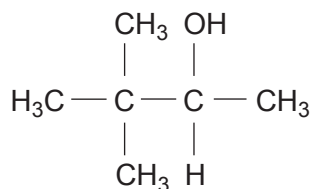


- (iv) Under suitable conditions pinacolone was reduced, giving a mixture of 2,2-dimethylbutane and 3,3-dimethylbutan-2-ol together with some unreacted pinacolone.

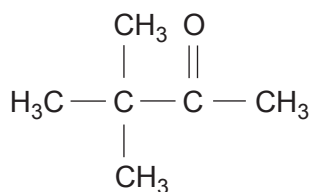
These three compounds were separated by gas chromatography.



2,2-dimethylbutane



3,3-dimethylbutan-2-ol



pinacolone

- I. Mary said that the boiling temperature of 3,3-dimethylbutan-2-ol would be the highest of these three compounds and that this could be used to identify it.

Explain why she is correct.

[2]

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- II. The ^{13}C NMR spectra of the remaining two compounds were taken.

Suggest how the ^{13}C NMR spectra of these two compounds would be similar and how they would be different.

[2]

Similar

.....

Different

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